Marcos Esparza

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Education

Bachelor of Science in Mechanical Engineering Expected December 2026
The University of Texas Permian Basin *Odessa, TX*
Relevant Coursework: Mechanical Design, Heat Transfer, Fluid Mechanics II; Engineering Materials & Optimization (in progress)
Activities: FAST, ASME, AIAA, SPE

Engineering Experience

Baker Hughes | Engineering & Technology Intern | Midland, TX | May 2026 – Present
ESP Performance Analysis
- Developed a preliminary methodology for evaluating returned electric submersible pump (ESP) performance across seven newer-versus-legacy product-family comparisons using shop PIT measurements of head, horsepower, efficiency, and vibration; evaluated field oil, water, and gas production information separately as operating context.
- Applied runtime-matched and geographically controlled comparisons, validated comparison outputs, reconciled engineering data, and presented findings, assumptions, and analytical limitations to product-line engineers.

ESP Design Intelligence
- Defined engineering requirements and comparison logic, developed and verified Python workflows, and created software tests for an application used by application engineers in design reviews; ongoing refinements incorporate engineering feedback.

Equipment Inspection & Failure Investigation
- Supported dismantle, inspection, and failure analysis (DIFA) of electric submersible pumps through hands-on component disassembly, cleaning, and inspection; reviewed evidence of wear and discussed potential failure mechanisms with engineers, customers, and consultants.

Engineering Projects

Falcon Aeronautics & Space Team (FAST) – 10,000-ft Competition Rocket
Aerodynamic Design Lead | Aug 2024 – Jun 2026; Mentor | Jun 2026 – Present
- Led aerodynamic design of modular aluminum fin-can assemblies used at IREC and the Lone Star Cup; developed geometry, SOLIDWORKS models, and manufacturing drawings and evaluated static stability and fin flutter using OpenRocket and MATLAB.
- Coordinated airframe integration with propulsion and recovery teams for telemetry-verified flights reaching approximately 8,800 ft and 9,300 ft; now mentor team members on rocket design and subsystem integration.

3D Printing Filament Extruder – Senior Design
Preliminary Design | Aug 2026 – Present
- Performed preliminary torque, power, gearing, extrusion-force, heater-sizing, heat-transfer, and shaft-loading calculations to guide motor selection; developed motor CAD in SOLIDWORKS and documented prototype manufacturing plans.

Thermo-Fluids / Mechanical Systems Lab – UTPB
Course Laboratory Projects | Fall 2025 – Spring 2026
- Tested centrifugal and gear pumps and a reciprocating compressor using dynamometer instrumentation and data acquisition; analyzed SR-30 turbojet performance using compressible-flow relations and a control-volume thrust model.

Technical Skills

Programming & Data: Python (pandas, NumPy, Matplotlib, SciPy, pytest); MATLAB (basic); Git/GitHub
Software: SOLIDWORKS, OpenRocket, Excel
Test & Lab: Pump and compressor testing, dynamometer instrumentation, pressure/flow measurement, data acquisition
Fabrication: Basic machining, hand tools, 3D printing, epoxy layups, rocket assembly and avionics integration
Certifications: Basin United; NAR High-Power Rocketry Level 1

Additional Experience

Logistics Associate | Academy Sports + Outdoors, Midland, TX | Jul 2024 – May 2026
Earlier Employment | H-E-B, Midland, TX | Oct 2022 – Jan 2024; Whataburger, Midland, TX | Oct 2020 – Mar 2022